COMPLETE MATHEMATICAL LECTURE NOTES ON NUMERICAL ANALYSIS

Exhaustive Step-by-Step Algebraic Derivations, Taylor Expansions, Error Proofs & Visual Graphs

Course Code: MATMJ54 — **Platform:** www.sciqb.com

Contents

1 Module 1: Errors and Their Computations

Numerical methods approximate continuous mathematical operations using discrete computational algorithms. A fundamental part of numerical analysis is evaluating, bounding, and minimizing errors.

1.1 Classification of Numerical Errors

1. **Inherent / Initial Errors:** Errors present in the raw input data prior to calculation due to physical measurement limits or empirical approximations.
2. **Truncation Errors:** Errors introduced when an infinite mathematical operation is approximated by a finite sequence of operations (e.g., truncating an infinite Taylor series).
3. **Round-off Errors:** Errors caused by representing real numbers with finite machine precision (floating-point representation) in computing systems.

1.2 Mathematical Definitions of Error

Definitions: Error Metrics

Let X denote the exact true value of a quantity and X^* denote its approximate value.

- **Absolute Error (E_A):**
$$E_A = |X - X^*| \tag{1}$$
- **Relative Error (E_R):**
$$E_R = \frac{|X - X^*|}{|X|} = \frac{E_A}{|X|}, \quad (X \neq 0) \tag{2}$$
- **Percentage Error (E_P):**
$$E_P = E_R \times 100\% = \frac{|X - X^*|}{|X|} \times 100\% \tag{3}$$

1.3 Significant Digits and Error Bounds

Theorem: Significant Digits Error Bound

An approximate number X^* is said to be correct to n decimal places (or significant digits) if the absolute error is bounded by:

$$E_A = |X - X^*| \leq \frac{1}{2} \times 10^{-n} \tag{4}$$

If X^* is rounded to n decimal places, the maximum truncation/rounding error is $\frac{1}{2} \cdot 10^{-n}$.

2 Module 2: Bisection and Regula-Falsi Methods

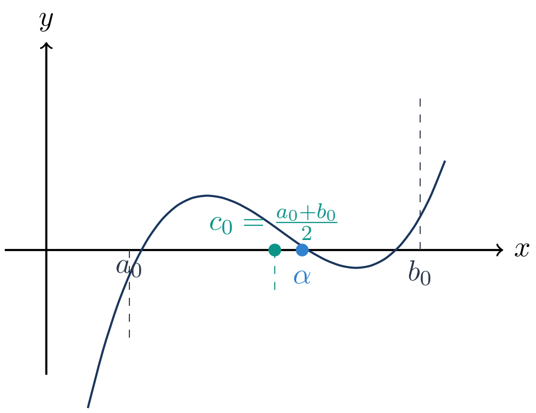
2.1 Intermediate Value Theorem (IVT)

Intermediate Value Theorem

If $f(x)$ is continuous on a closed interval $[a, b]$ and $f(a) \cdot f(b) < 0$ (meaning $f(a)$ and $f(b)$ have opposite signs), then there exists at least one root $\alpha \in (a, b)$ such that $f(\alpha) = 0$.

2.2 Bisection Method

The Bisection Method repeatedly halves the bracket interval $[a, b]$.



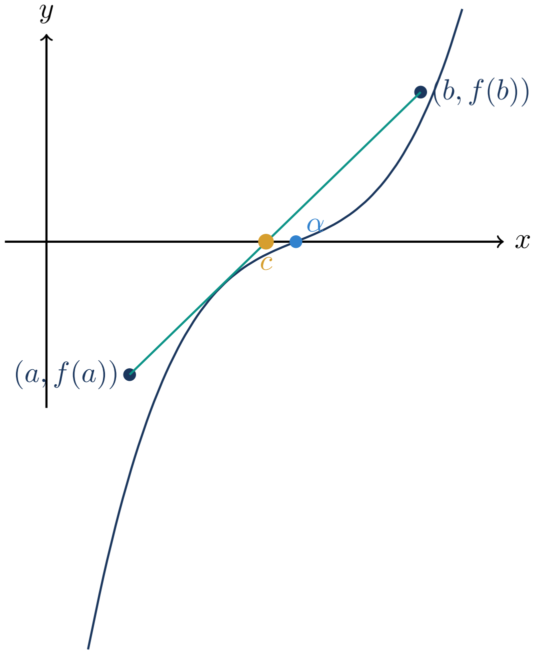
Bisection Iteration Formula and Error Bound Proof

Midpoint formula: $c_k = \frac{a_k+b_k}{2}$.
Complete Step-by-Step Proof of Error Bound $|c_n - \alpha| \leq \frac{b-a}{2^n}$:

1. **Initial Step ($n = 0$):** The initial interval is $[a_0, b_0]$ with length $b_0 - a_0 = b - a$.
2. **Interval Halving after n Steps:** At each successive iteration, the interval is halved:
$$\text{Length of } [a_n, b_n] = b_n - a_n = \frac{b-a}{2^n}$$
3. **Root Proximity to Midpoint:** Since $f(a_n)f(b_n) < 0$, the exact root α lies inside $[a_n, b_n]$. The midpoint $c_n = \frac{a_n+b_n}{2}$ is at the center of $[a_n, b_n]$. Thus, the distance between c_n and any point in $[a_n, b_n]$ cannot exceed half the interval length:
$$|c_n - \alpha| \leq \frac{b_n - a_n}{2} = \frac{b-a}{2^{n+1}} \leq \frac{b-a}{2^n} \qquad \square \tag{5}$$

Rate of Convergence: The Bisection method converges **linearly** with order $p = 1$ and asymptotic error constant $C = 1/2$.

2.3 Regula-Falsi (False Position) Method



Regula-Falsi Formula

The secant line connecting $(a, f(a))$ and $(b, f(b))$ has equation:

$$y - f(a) = \frac{f(b) - f(a)}{b - a}(x - a) \tag{6}$$

Setting $y = 0$ gives the root approximation c:

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)} = a - \frac{b - a}{f(b) - f(a)} f(a) \tag{7}$$

2.3.1 Exhaustive Step-by-Step Proof of Regula-Falsi Linear Convergence

Setting: Let α be the exact root ($f(\alpha) = 0$). In the Regula-Falsi method, one of the bracket endpoints remains fixed (say $x_0 = a = \alpha + e_0$, where e_0 is the constant error of the fixed endpoint), while the other endpoint $x_k = \alpha + e_k$ moves towards α.

Step 1: Set up the Error Recurrence Equation

Substitute $x_k = \alpha + e_k$, $x_0 = \alpha + e_0$, and $x_{k+1} = \alpha + e_{k+1}$ into the iteration formula:

$$\begin{aligned}
x_{k+1} &= x_k - \frac{x_k - x_0}{f(x_k) - f(x_0)} f(x_k) \\
\alpha + e_{k+1} &= (\alpha + e_k) - \frac{(\alpha + e_k) - (\alpha + e_0)}{f(\alpha + e_k) - f(\alpha + e_0)} f(\alpha + e_k) \\
e_{k+1} &= e_k - (e_k - e_0) \frac{f(\alpha + e_k)}{f(\alpha + e_k) - f(\alpha + e_0)}
\end{aligned} \tag{8}$$

Step 2: Taylor Series Expansions about the Exact Root α

Since $f(\alpha) = 0$, expanding $f(\alpha + e_k)$ and $f(\alpha + e_0)$ in Taylor series gives:

$$\begin{aligned} f(\alpha + e_k) &= e_k f'(\alpha) + \frac{e_k^2}{2!} f''(\alpha) + \frac{e_k^3}{3!} f'''(\alpha) + \dots \\ &= e_k f'(\alpha) \left[1 + \frac{1}{2} e_k \frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6} e_k^2 \frac{f'''(\alpha)}{f'(\alpha)} + \dots \right] \end{aligned} \tag{9}$$

Similarly for the fixed endpoint:

$$f(\alpha + e_0) = e_0 f'(\alpha) + \frac{e_0^2}{2!} f''(\alpha) + \frac{e_0^3}{3!} f'''(\alpha) + \dots \tag{10}$$

Subtracting $f(\alpha + e_0)$ from $f(\alpha + e_k)$ for the denominator:

$$\begin{aligned} f(\alpha + e_k) - f(\alpha + e_0) &= (e_k - e_0) f'(\alpha) + \frac{e_k^2 - e_0^2}{2!} f''(\alpha) + \frac{e_k^3 - e_0^3}{3!} f'''(\alpha) + \dots \\ &= (e_k - e_0) f'(\alpha) \left[1 + \frac{1}{2}(e_k + e_0) \frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}(e_k^2 + e_k e_0 + e_0^2) \frac{f'''(\alpha)}{f'(\alpha)} + \dots \right] \end{aligned} \tag{11}$$

Step 3: Canceling Common Factors and Applying Binomial Expansion

Substitute equations (9) and (11) into the error formula:

$$
\begin{aligned}
e_{k+1} &= e_k - (e_k - e_0)\frac{e_k f'(\alpha)\left[1 + \frac{1}{2}e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right]}{(e_k - e_0)f'(\alpha)\left[1 + \frac{1}{2}(e_k + e_0)\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}(e_k^2 + e_k e_0 + e_0^2)\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right]} \\
&= e_k - e_k\left[1 + \frac{1}{2}e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right] \\
&\quad \times \left[1 + \frac{1}{2}(e_k + e_0)\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}(e_k^2 + e_k e_0 + e_0^2)\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right]^{-1} \qquad (12)
\end{aligned}
$$

Using the Binomial expansion identity $(1+u)^{-1} = 1 - u + u^2 - \dots$:

$$
\begin{aligned}
&\left[1 + \frac{1}{2}(e_k + e_0)\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}(e_k^2 + e_k e_0 + e_0^2)\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right]^{-1} \\
&= 1 - \left\{\frac{1}{2}(e_k + e_0)\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}(e_k^2 + e_k e_0 + e_0^2)\frac{f'''(\alpha)}{f'(\alpha)}\right\} + \left\{\frac{1}{2}(e_k + e_0)\frac{f''(\alpha)}{f'(\alpha)}\right\}^2 - \dots \qquad (13)
\end{aligned}
$$

Step 4: Term-by-Term Cross Multiplication and Simplification

Multiplying the two bracketed series:

$$
\begin{aligned}
&= 1 + \frac{1}{2}e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} - \frac{1}{2}(e_k + e_0)\frac{f''(\alpha)}{f'(\alpha)} - \frac{1}{6}(e_k^2 + e_k e_0 + e_0^2)\frac{f'''(\alpha)}{f'(\alpha)} \\
&\quad - \frac{1}{4}e_k(e_k + e_0)\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 + \frac{1}{4}(e_k^2 + 2e_k e_0 + e_0^2)\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 + \dots \\
&= 1 + \frac{1}{2}\{e_k - (e_k + e_0)\}\frac{f''(\alpha)}{f'(\alpha)} + (e_k e_0 + e_0^2)\left[\frac{1}{4}\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 - \frac{1}{6}\frac{f'''(\alpha)}{f'(\alpha)}\right] + \dots \\
&= 1 - \frac{1}{2}e_0\frac{f''(\alpha)}{f'(\alpha)} + (e_k e_0 + e_0^2)\left[\frac{1}{4}\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 - \frac{1}{6}\frac{f'''(\alpha)}{f'(\alpha)}\right] + \dots \qquad (14)
\end{aligned}
$$

Substitute this expression back into the equation for e_{k+1}:

$$
\begin{aligned}
e_{k+1} &= e_k - e_k\left[1 - \frac{1}{2}e_0\frac{f''(\alpha)}{f'(\alpha)} + (e_k e_0 + e_0^2)\left\{\frac{1}{4}\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 - \frac{1}{6}\frac{f'''(\alpha)}{f'(\alpha)}\right\} + \dots\right] \\
&= e_k - e_k + e_k e_0\frac{f''(\alpha)}{2f'(\alpha)} + (e_k^2 e_0 + e_k e_0^2)\left[\frac{1}{6}\frac{f'''(\alpha)}{f'(\alpha)} - \frac{1}{4}\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2\right] + \dots \\
\mathbf{e_{k+1}} &= \mathbf{e_k e_0}\frac{\mathbf{f''}(\alpha)}{\mathbf{2f'}(\alpha)} + \mathbf{O}(\mathbf{e_k^2 e_0} + \mathbf{e_k e_0^2}) \qquad (15)
\end{aligned}
$$

Step 5: Order of Convergence Conclusion

Since the initial endpoint $x_0 = a$ is fixed throughout the iterations, e_0 is a non-zero constant. Defining $A = \frac{e_0 f''(\alpha)}{2f'(\alpha)}$:

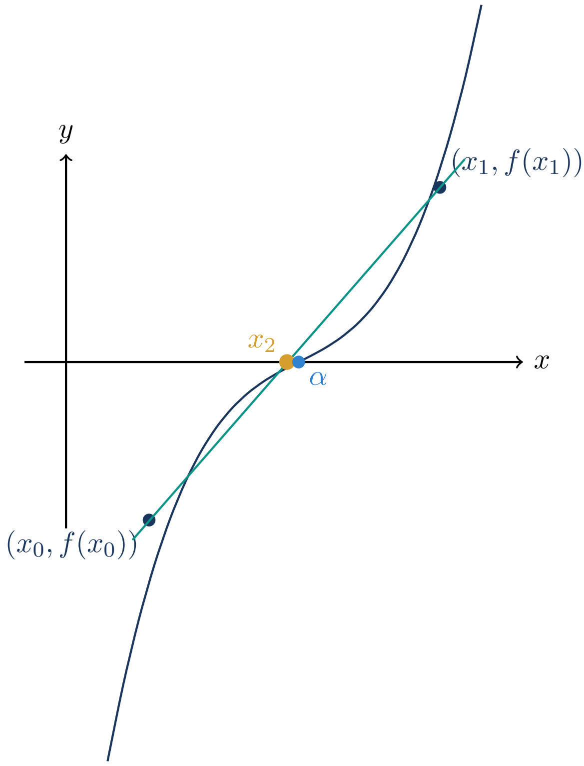
$$e_{k+1} = Ae_k \tag{16}$$

Comparing with the general error relation $|e_{k+1}| = C|e_k|^p$ gives $p = 1$.

Rate of Convergence of Regula-Falsi Method is $\mathbf{p = 1}$ (Linear Convergence) $\square$

3 Module 3: Secant Method and Rate of Convergence

3.1 Derivation and Iteration Formula



Secant Method Iteration

Unlike Regula-Falsi, the Secant Method does not require bracketing $f(x_{k-1})f(x_k) < 0$. It draws a secant line through the two most recent iterates $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$:

$$x_{k+1} = x_k - \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} f(x_k), \quad k = 1, 2, 3, \dots \tag{17}$$

3.2 Exhaustive Proof: Rate of Convergence of Secant Method ($p \approx 1.6180$)

Step 1 to 3: Error Recurrence and Asymptotic Form

Step 1: Write Error Recurrence Relation Let $x_{k-1} = \alpha + e_{k-1}, x_k = \alpha + e_k, x_{k+1} = \alpha + e_{k+1}$:

$$e_{k+1} = e_k - (e_k - e_{k-1})\frac{f(\alpha + e_k)}{f(\alpha + e_k) - f(\alpha + e_{k-1})} \tag{18}$$

Step 2: Taylor Expansion & Asymptotic Error Relation Following the exact algebraic simplification as derived in Module 2, but with the moving iterate e_{k-1} in place of the fixed e_0:

$$e_{k+1} = e_k e_{k-1}\frac{f''(\alpha)}{2f'(\alpha)} + O(e_k^2 e_{k-1} + e_k e_{k-1}^2) \tag{19}$$

Defining asymptotic error constant $A = \frac{f''(\alpha)}{2f'(\alpha)}$:

$$e_{k+1} \approx A e_k e_{k-1} \quad \text{— (Equation 1)} \tag{20}$$

Step 3: Postulate the Order of Convergence Relation Assume an error relation of the form:

$$e_{k+1} = c e_k^p \quad \text{— (Equation 2)} \tag{21}$$

where c is a constant and p is the order of convergence. Expressing e_k and e_{k-1} from Equation (2):

$$\begin{aligned} e_k &= c^{-1/p} e_{k+1}^{1/p} \\ e_{k-1} &= c^{-1/p} e_k^{1/p} \quad \text{— (Equation 3)} \end{aligned} \tag{22}$$

Step 4 to 5: Characteristic Exponent Equation and Root Calculation

Step 4: Substitute Equations (2) and (3) into Equation (1)

$$\begin{aligned} c e_k^p &= A e_k \left(c^{-1/p} e_k^{1/p}\right) \\ c^{1+1/p} e_k^p &= A e_k^{1+1/p} \end{aligned} \tag{23}$$

Step 5: Equate Exponents of e_k and Solve the Characteristic Equation Equating the exponents of e_k on both sides:

$$\begin{aligned} p &= 1 + \frac{1}{p} \\ p^2 - p - 1 &= 0 \\ p &= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)} = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2} \end{aligned} \tag{24}$$

Evaluating the roots:

- $p_1 = \frac{1-\sqrt{5}}{2} = \frac{1-2.236068}{2} \approx -0.61803$ (Discard negative root)
- $p_2 = \frac{1+\sqrt{5}}{2} = \frac{1+2.236068}{2} \approx \mathbf{1.61803}$

Rate of Convergence of Secant Method is $\mathbf{p} = \mathbf{\frac{1+\sqrt{5}}{2}} \approx \mathbf{1.6180}$ (Superlinear / Golden Ratio) □

3.3 Solved Example: Regula-Falsi vs. Secant Method

Comparative Worked Example: $f(x) = x^3 + x^2 - 1 = 0$ on $[0, 1]$

Observe $f(0) = 0^3 + 0^2 - 1 = -1 < 0$ and $f(1) = 1^3 + 1^2 - 1 = 1 > 0$. The root lies in $[0, 1]$.

1. Regula-Falsi Method ($a = 0, b = 1$):

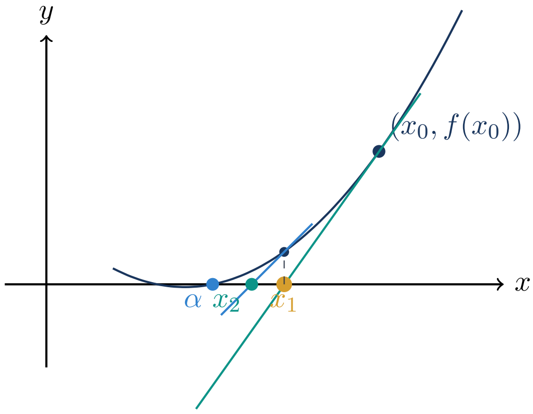
- $k = 1$: $c = \frac{0(1)-1(-1)}{1-(-1)} = \frac{1}{2} = 0.5$. $f(0.5) = 0.5^3 + 0.5^2 - 1 = -0.625 < 0 \implies [0.5, 1]$
- $k = 2$: $c = \frac{0.5(1)-1(-0.625)}{1-(-0.625)} = \frac{1.125}{1.625} \approx 0.692307$. $f(0.692307) = -0.188916 \implies [0.692307, 1]$
- $k = 3$: $c = \frac{0.692307(1)-1(-0.188916)}{1-(-0.188916)} = \frac{0.881223}{1.188916} \approx 0.741198$. $f(0.741198) = -0.043430 \implies [0.741198, 1]$
- $k = 4$: $c = \frac{0.741198(1)-1(-0.043430)}{1+0.043430} \approx 0.751969$. $f(0.751969) = -0.009336 \implies [0.751969, 1]$
- $k = 5$: $c = \frac{0.751969(1)-1(-0.009336)}{1+0.009336} \approx \mathbf{0.754263}$.

2. Secant Method ($x_0 = 0, x_1 = 1$):

- $x_0 = 0, f(x_0) = -1$
- $x_1 = 1, f(x_1) = 1$
- $x_2 = 1 - \frac{1-0}{1-(-1)}(1) = 1 - 0.5 = 0.5, \quad f(0.5) = -0.625$
- $x_3 = 0.5 - \frac{0.5-1}{-0.625-1}(-0.625) = 0.5 + \frac{-0.5}{-1.625}(-0.625) = 0.692307, \quad f(x_3) = -0.188916$
- $x_4 = 0.692307 - \frac{0.692307-0.5}{-0.188916-(-0.625)}(-0.188916) = 0.692307 + \frac{0.192307(0.188916)}{0.436084} \approx 0.775607, \quad f(x_4) = 0.066472$
- $x_5 = 0.775607 - \frac{0.775607-0.692307}{0.066472-(-0.188916)}(0.066472) \approx \mathbf{0.753951}$.

4 Module 4: Newton-Raphson Method and Applications

4.1 Derivation and Tangent Formula



Newton-Raphson Formula

Let x_0 be an initial guess for the root of $f(x) = 0$. The tangent line to $y = f(x)$ at $(x_0, f(x_0))$ is:

$$y - f(x_0) = f'(x_0)(x - x_0) \tag{25}$$

Setting $y = 0$ gives the first approximation x_1:

$$0 - f(x_0) = f'(x_0)(x_1 - x_0) \implies x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \tag{26}$$

Iterating k times gives the general Newton-Raphson formula:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad f'(x_k) \neq 0 \tag{27}$$

4.2 Exhaustive Proof: Quadratic Convergence ($p = 2$)

Step 1: Set up Error Expression from Iteration Formula

Let α be the exact root such that $f(\alpha) = 0$. Let $x_k = \alpha + e_k$ and $x_{k+1} = \alpha + e_{k+1}$, where e_k and e_{k+1} are the errors at steps k and $k + 1$:

$$\begin{aligned} \alpha + e_{k+1} &= \alpha + e_k - \left[\frac{f(\alpha + e_k)}{f'(\alpha + e_k)}\right] \\ e_{k+1} &= e_k - \frac{f(\alpha + e_k)}{f'(\alpha + e_k)} \end{aligned} \tag{28}$$

Step 2: Taylor Expansions of $f(\alpha + e_k)$ and $f'(\alpha + e_k)$ up to 3rd Derivative

Expanding $f(\alpha + e_k)$ and $f'(\alpha + e_k)$ about α in Taylor series:

$$\begin{aligned} f(\alpha + e_k) &= f(\alpha) + e_k f'(\alpha) + \frac{e_k^2}{2!} f''(\alpha) + \frac{e_k^3}{3!} f'''(\alpha) + \dots \\ f'(\alpha + e_k) &= f'(\alpha) + e_k f''(\alpha) + \frac{e_k^2}{2!} f'''(\alpha) + \dots \end{aligned} \tag{29}$$

Since α is a root of $f(x) = 0$, substitute $f(\alpha) = 0$:

$$e_{k+1} = e_k - \frac{\left[e_k f'(\alpha) + \frac{e_k^2}{2!} f''(\alpha) + \frac{e_k^3}{3!} f'''(\alpha) + \dots\right]}{\left[f'(\alpha) + e_k f''(\alpha) + \frac{1}{2} e_k^2 f'''(\alpha) + \dots\right]} \tag{30}$$

Step 3: Factoring $f'(\alpha)$ out of Numerator and Denominator

Factor out $e_k f'(\alpha)$ from the numerator and $f'(\alpha)$ from the denominator:

$$e_{k+1} = e_k - \frac{e_k f'(\alpha)\left[1 + \frac{e_k}{2!}\frac{f''(\alpha)}{f'(\alpha)} + \frac{e_k^2}{3!}\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right]}{f'(\alpha)\left[1 + \left\{e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{2}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right\}\right]}$$
$$= e_k - e_k\left[1 + \frac{1}{2}e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right] \times \left[1 + \left\{e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{2}e_k^2\frac{f'''(\alpha)}{f'(\alpha)}\right\}\right]^{-1} \quad (31)$$

Step 4: Applying Binomial Series Expansion $(1+u)^{-1} = 1 - u + u^2 - \dots$

Expanding the inverted denominator using the binomial expansion:

$$\left[1 + \left\{e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{2}e_k^2\frac{f'''(\alpha)}{f'(\alpha)}\right\}\right]^{-1}$$
$$= 1 - \left\{e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{2}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right\} + \left\{e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{2}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right\}^2 - \dots$$
$$= 1 - e_k\frac{f''(\alpha)}{f'(\alpha)} - \frac{1}{2}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + e_k^2\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 - \dots \quad (32)$$

Step 5: Full Term-by-Term Cross Multiplication

Multiplying the two series expansion brackets together:

$$= \left[1 + \frac{1}{2}e_k\frac{f''(\alpha)}{f'(\alpha)} + \frac{1}{6}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + \dots\right] \times \left[1 - e_k\frac{f''(\alpha)}{f'(\alpha)} - \frac{1}{2}e_k^2\frac{f'''(\alpha)}{f'(\alpha)} + e_k^2\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 - \dots\right]$$
$$= 1 + e_k\left\{\frac{1}{2} - 1\right\}\frac{f''(\alpha)}{f'(\alpha)} + e_k^2\left\{\frac{1}{6}\frac{f'''(\alpha)}{f'(\alpha)} - \frac{1}{2}\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 - \frac{1}{2}\frac{f'''(\alpha)}{f'(\alpha)} + \left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2\right\} + \dots$$
$$= 1 - \frac{1}{2}e_k\frac{f''(\alpha)}{f'(\alpha)} + e_k^2\left\{\left(\frac{1}{6} - \frac{1}{2}\right)\frac{f'''(\alpha)}{f'(\alpha)} + \left(1 - \frac{1}{2}\right)\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2\right\} + \dots$$
$$= 1 - \frac{1}{2}e_k\frac{f''(\alpha)}{f'(\alpha)} + e_k^2\left\{-\frac{1}{3}\frac{f'''(\alpha)}{f'(\alpha)} + \frac{1}{2}\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2\right\} + \dots \quad (33)$$

Step 6: Substituting into e_{k+1} and Final Order of Convergence

Substituting the product back into the e_{k+1} equation:

$$\begin{aligned}
e_{k+1} &= e_k - e_k \left[1 - \frac{1}{2} e_k \frac{f''(\alpha)}{f'(\alpha)} + e_k^2 \left\{ -\frac{1}{3} \frac{f'''(\alpha)}{f'(\alpha)} + \frac{1}{2} \left(\frac{f''(\alpha)}{f'(\alpha)} \right)^2 \right\} + \dots \right] \\
&= e_k - e_k + \frac{e_k^2}{2} \frac{f''(\alpha)}{f'(\alpha)} + e_k^3 \left\{ \frac{1}{3} \frac{f'''(\alpha)}{f'(\alpha)} - \frac{1}{2} \left(\frac{f''(\alpha)}{f'(\alpha)} \right)^2 \right\} + \dots \\
&= \frac{f''(\alpha)}{2f'(\alpha)} e_k^2 + e_k^3 \left\{ \frac{1}{3} \frac{f'''(\alpha)}{f'(\alpha)} - \frac{1}{2} \left(\frac{f''(\alpha)}{f'(\alpha)} \right)^2 \right\} + \dots \\
\mathbf{e_{k+1}} &= \frac{\mathbf{f''}(\alpha)}{\mathbf{2f'}(\alpha)} \mathbf{e_k^2} + \mathbf{O}(\mathbf{e_k^3}) \qquad (34)
\end{aligned}$$

Defining asymptotic error constant $C = \left| \frac{f''(\alpha)}{2f'(\alpha)} \right|$:

$$|e_{k+1}| \le C|e_k|^2 \implies \mathbf{p = 2} \quad \text{(Quadratic Order of Convergence)} \quad \square \qquad (35)$$

4.3 Solved Newton-Raphson Polynomial Examples

Example 1: $f(x) = x^3 + x^2 - 1 = 0$

Given $f(x) = x^3 + x^2 - 1 \implies f'(x) = 3x^2 + 2x$. Iteration formula:

$$x_{k+1} = x_k - \frac{x_k^3 + x_k^2 - 1}{3x_k^2 + 2x_k} = \frac{3x_k^3 + 2x_k^2 - x_k^3 - x_k^2 + 1}{3x_k^2 + 2x_k} = \frac{\mathbf{2x_k^3 + x_k^2 + 1}}{\mathbf{3x_k^2 + 2x_k}} \tag{36}$$

Let initial approximation $x_0 = 1$:

- $k = 0$: $x_1 = \frac{2(1)^3+1^2+1}{3(1)^2+2(1)} = \frac{4}{5} = \mathbf{0.8}$
- $k = 1$: $x_2 = \frac{2(0.8)^3+0.8^2+1}{3(0.8)^2+2(0.8)} = \frac{2(0.512)+0.64+1}{3(0.64)+1.6} = \frac{1.024+0.64+1}{1.92+1.6} = \frac{2.664}{3.52} \approx \mathbf{0.756818}$
- $k = 2$: $x_3 = \frac{2(0.756818)^3+(0.756818)^2+1}{3(0.756818)^2+2(0.756818)} \approx \mathbf{0.754878}$

Practice Problems:

1. $x^3 - 2x - 5 = 0$
2. $x^3 + 4x^2 - 3x + 1 = 0$
3. $x^3 - 5x + 3 = 0$

4.4 Special Newton-Raphson Algorithms

1. Cube Root Algorithm ($\sqrt[3]{N}$)

Let $x = \sqrt[3]{N} \implies x^3 = N \implies f(x) = x^3 - N = 0, \quad f'(x) = 3x^2$. Newton-Raphson formula:

$$x_{k+1} = x_k - \frac{x_k^3 - N}{3x_k^2} = \frac{3x_k^3 - x_k^3 + N}{3x_k^2} = \frac{\mathbf{2x_k^3 + N}}{\mathbf{3x_k^2}} \tag{37}$$

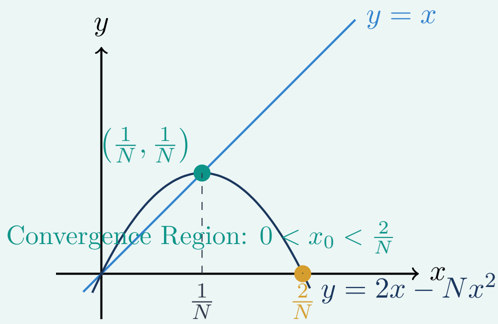
Numerical Example: Find $\sqrt[3]{37}$ using Newton-Raphson

- Let $f(x) = x^3 - 37 = 0$.
- Test integers: $f(3) = 3^3 - 37 = 27 - 37 = -10 < 0$ and $f(4) = 4^3 - 37 = 64 - 37 = 27 > 0$.
- Root lies in $[3, 4]$. Choose initial guess $x_0 = 3$.
- Iteration formula: $x_{k+1} = \frac{2x_k^3+37}{3x_k^2}$.
- $k = 0$: $x_1 = \frac{2(3)^3+37}{3(3)^2} = \frac{2(27)+37}{3(9)} = \frac{54+37}{27} = \frac{91}{27} \approx \mathbf{3.370370}$
- $k = 1$: $x_2 = \frac{2(3.370370)^3+37}{3(3.370370)^2} = \frac{2(38.2854)+37}{3(11.3594)} = \frac{113.5708}{34.0782} \approx \mathbf{3.332611}$

2. Division-Free Reciprocal Algorithm ($1/N$) and Full Convergence Range Proof

To find the reciprocal $x = 1/N$ without performing hardware division: Let $f(x) = \frac{1}{x} - N = 0 \implies f'(x) = -\frac{1}{x^2}$.

$$x_{k+1} = x_k - \frac{\frac{1}{x_k} - N}{-\frac{1}{x_k^2}} = x_k + x_k^2\left(\frac{1}{x_k} - N\right) = x_k + x_k - Nx_k^2 = \mathbf{2x_k - Nx_k^2} \tag{38}$$



Geometric Analysis of Iteration Function: The iteration curve is the downward parabola $y = 2x - Nx^2$:

$$y = -N\left(x^2 - \frac{2x}{N}\right) = -N\left[\left(x - \frac{1}{N}\right)^2 - \frac{1}{N^2}\right] \implies y - \frac{1}{N} = -N\left(x - \frac{1}{N}\right)^2$$

The vertex is located at $\left(\frac{1}{N}, \frac{1}{N}\right)$ and the parabola crosses the x-axis at $x = 0$ and $x = \frac{2}{N}$.
Full Algebraic Proof that Convergence Requires $0 < x_0 < \frac{2}{N}$:

1. Define error at iteration $k+1$: $e_{k+1} = \frac{1}{N} - x_{k+1}$.

$$\begin{aligned} e_{k+1} &= \frac{1}{N} - (2x_k - Nx_k^2) = \frac{1}{N} - 2x_k + Nx_k^2 \\ &= N\left(\frac{1}{N^2} - \frac{2x_k}{N} + x_k^2\right) = N\left(\frac{1}{N} - x_k\right)^2 = \mathbf{Ne_k^2} \end{aligned} \tag{39}$$

2. Multiplying both sides by N:

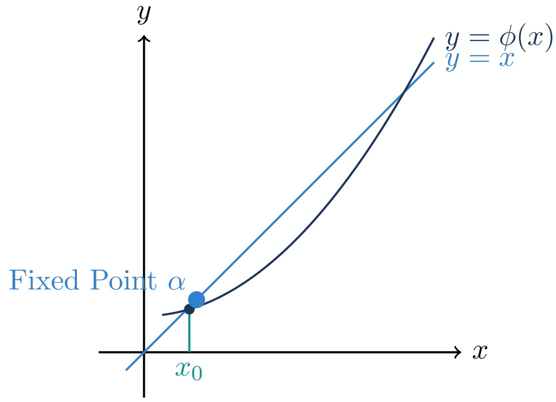
$$Ne_{k+1} = (Ne_k)^2 = (Ne_{k-1})^4 = \cdots = (Ne_0)^{2^{k+1}} \tag{40}$$

3. For the error sequence to converge to zero ($e_k \to 0$ as $k \to \infty$), the base must satisfy:

$$\begin{aligned} |Ne_0| < 1 &\iff \left|N\left(\frac{1}{N} - x_0\right)\right| < 1 \\ &\iff |1 - Nx_0| < 1 \\ &\iff -1 < 1 - Nx_0 < 1 \\ &\iff -2 < -Nx_0 < 0 \\ &\iff \mathbf{0 < x_0 < \frac{2}{N}} \qquad \square \end{aligned} \tag{41}$$

5 Module 5: Fixed-Point Iteration Method

5.1 Formulation and Cobweb Diagram



Fixed-Point Iteration Scheme

Given an equation $f(x) = 0$, rewrite it in the equivalent fixed-point form:

$$x = \phi(x) \tag{42}$$

Starting from an initial guess x_0, generate successive approximations:

$$\begin{aligned}
\text{First approximation:} \quad & x_1 = \phi(x_0) \\
\text{Second approximation:} \quad & x_2 = \phi(x_1) \\
\text{Third approximation:} \quad & x_3 = \phi(x_2) \\
& \vdots \\
n\text{-th approximation:} \quad & x_n = \phi(x_{n-1}), \quad n = 1, 2, 3, \dots
\end{aligned} \tag{43}$$

5.2 Exhaustive Step-by-Step Proof: Fixed-Point Convergence Theorem

Theorem Statement: Fixed-Point Iteration Convergence

Theorem: If

(i) α be a root of $f(x) = 0$, which is equivalent to $x = \phi(x)$.

(ii) I be any interval containing the root point $x = \alpha$.

(iii) $|\phi'(x)| < 1$ for all $x \in I$.

Then the sequence of iterative approximations $x_0, x_1, x_2, \dots, x_n, \dots$ will converge to the root α, provided the initial approximation x_0 is chosen in I.

Step 1 to 3: Setting Up Successive Differences

Step 1: Root Condition Since α is the exact root, it satisfies the fixed-point equation $x = \phi(x)$:

$$\alpha = \phi(\alpha) \quad \text{— (1)} \tag{44}$$

Step 2: Iteration Condition If x_{n-1} and x_n be two successive approximations to the root α, then:

$$x_n = \phi(x_{n-1}) \quad \text{— (2)} \tag{45}$$

Step 3: Subtract Equation (1) from Equation (2) Subtracting (1) from (2) gives:

$$x_n - \alpha = \phi(x_{n-1}) - \phi(\alpha) \quad \text{— (3)} \tag{46}$$

Step 4 to 6: Mean Value Theorem Application and Contraction Inequality

Step 4: Application of Lagrange's Mean Value Theorem By Lagrange's Mean Value Theorem, there exists a point ξ strictly between x_{n-1} and α such that:

$$\frac{\phi(x_{n-1}) - \phi(\alpha)}{x_{n-1} - \alpha} = \phi'(\xi) \implies \phi(x_{n-1}) - \phi(\alpha) = (x_{n-1} - \alpha)\phi'(\xi) \tag{47}$$

Step 5: Substitute into Equation (3)

$$x_n - \alpha = (x_{n-1} - \alpha)\phi'(\xi) \tag{48}$$

Taking the absolute value on both sides:

$$|x_n - \alpha| = |x_{n-1} - \alpha| \cdot |\phi'(\xi)| \quad \text{— (4)} \tag{49}$$

Step 6: Bound by Contraction Factor k Given that $|\phi'(x)| \leq k < 1$ for all $x \in I$, and since $\xi \in I$, we have $|\phi'(\xi)| \leq k$:

$$|x_n - \alpha| \leq k|x_{n-1} - \alpha| \quad \text{— (5)} \tag{50}$$

Step 7 to 9: Backward Induction, Limit as $n \to \infty$ and Convergence Rate

Step 7: Recursive Substitution (Backward Induction) In Equation (5), replace n by $n-1$:

$$|x_{n-1} - \alpha| \leq k|x_{n-2} - \alpha| \tag{51}$$

Substituting this back into (5):

$$|x_n - \alpha| \leq k \cdot [k|x_{n-2} - \alpha|] = k^2|x_{n-2} - \alpha| \tag{52}$$

Similarly, continuing this process recursively for $n-2, n-3, \ldots, 0$:

$$|x_n - \alpha| \leq k^n|x_0 - \alpha| \tag{53}$$

Step 8: Taking Limit as $n \to \infty$ Since $0 \leq k < 1$, as $n \to \infty$, $k^n \to 0$:

$$\lim_{n\to\infty} |x_n - \alpha| \leq |x_0 - \alpha| \lim_{n\to\infty} k^n = 0 \tag{54}$$

$$\Rightarrow \lim_{n\to\infty} x_n = \alpha \tag{55}$$

Therefore, the sequence of approximations converges towards the exact root α.

Step 9: Rate of Convergence Let e_n and e_{n-1} be the errors in the n-th and $(n-1)$-th iterations:

$$e_n = x_n - \alpha, \quad e_{n-1} = x_{n-1} - \alpha$$

From Equation (5):

$$|e_n| \leq k|e_{n-1}| \tag{56}$$

Comparing with standard error recurrence $|e_n| = C|e_{n-1}|^p$:

Rate of Convergence of Fixed-Point Iteration is $\mathbf{p = 1}$ (Linear Convergence) $\square$

5.3 Solved Fixed-Point Iteration Examples

Example 1: Solve $x^3 - 3x + 1 = 0$ on $[0, 1]$

Step 1: Check Interval Signs $f(0) = 0^3 - 3(0) + 1 = 1 > 0$ and $f(1) = 1^3 - 3(1) + 1 = -1 < 0$. Root lies in $[0, 1]$.
Step 2: Formulate $\phi(x)$ and Check Convergence Condition Rewrite $x^3 - 3x + 1 = 0 \implies 3x = x^3 + 1 \implies x = \phi(x) = \frac{1}{3}(x^3 + 1)$. Compute derivative:

$$\phi'(x) = \frac{1}{3}(3x^2) = x^2$$

On $[0, 1)$, $|\phi'(x)| = x^2 < 1$. Thus, the iteration will converge!
Step 3: Perform Iterations starting from $x_0 = 0$

- $n = 1$: $x_1 = \phi(x_0) = \frac{1}{3}(0^3 + 1) = \frac{1}{3} \approx \mathbf{0.333333}$
- $n = 2$: $x_2 = \phi(x_1) = \frac{1}{3}((0.333333)^3 + 1) = \frac{1}{3}(0.037037 + 1) = \frac{1.037037}{3} \approx \mathbf{0.345679}$
- $n = 3$: $x_3 = \phi(x_2) = \frac{1}{3}((0.345679)^3 + 1) = \frac{1}{3}(0.041306 + 1) = \frac{1.041306}{3} \approx \mathbf{0.347098}$

Example 2: Determine Parameter K for $x^3 - 15 = 0$ on $[2, 3]$

Problem Statement: Solve $x^3 - 15 = 0$ on $[2, 3]$ using the iteration form $x = \phi(x) = x + \frac{1}{K}(15 - x^3)$. Find an integer K ensuring convergence.
Step 1: Check Interval Signs $f(2) = 2^3 - 15 = 8 - 15 = -7 < 0$ and $f(3) = 3^3 - 15 = 27 - 15 = 12 > 0$. Root lies in $[2, 3]$.
Step 2: Compute Derivative $\phi'(x)$ and Set Convergence Condition

$$\phi(x) = x + \frac{15 - x^3}{K} \implies \phi'(x) = 1 - \frac{3x^2}{K}$$

Convergence requires $|\phi'(x)| < 1$ for all $x \in [2, 3]$:

$$-1 < 1 - \frac{3x^2}{K} < 1$$

Subtract 1 from all parts:

$$-2 < -\frac{3x^2}{K} < 0 \implies 0 < \frac{3x^2}{K} < 2 \implies x^2 < \frac{2K}{3}$$

Step 3: Bound x^2 on the Interval $[2, 3]$ Since x^2 is a strictly increasing function on $[2, 3]$, its maximum occurs at $x = 3$:

$$\max_{x\in[2,3]} x^2 = 3^2 = 9$$

Therefore, to satisfy $x^2 < \frac{2K}{3}$ for all $x \in [2, 3]$:

$$9 < \frac{2K}{3} \implies 2K > 27 \implies K > \frac{27}{2} = 13.5$$

We choose the smallest convenient integer: $\mathbf{K = 15} \geq 14$.
Step 4: Execute Iterations with $K = 15$ and $x_0 = 2$ Iteration formula: $\phi(x) = x + \frac{15-x^3}{15} = \frac{15x+15-x^3}{15}$.

- $x_0 = 2$
- $x_1 = \phi(x_0) = \frac{15(2)+15-2^3}{15} = \frac{30+15-8}{15} = \frac{37}{15} \approx \mathbf{2.466667}$
- $x_2 = \phi(x_1) = \frac{15(2.466667)+15-(2.466667)^3}{15} = \frac{37+15-15.0084}{15} = \frac{36.9916}{15} \approx \mathbf{2.466107}$

6 Module 6: Master Quick Revision Cheat Sheet

1. Numerical Root-Finding Methods Comparison Table

- **Bisection Method (Interval Halving):**
  - **Formula:** $x_{n+1} = \frac{a_n+b_n}{2}$ with bracket condition $f(a)f(b) < 0$.
  - **Order of Convergence (p): $\mathbf{p = 1}$** (Linear).
  - **Error Relation:** $|e_n| = \frac{b-a}{2^n}$.
  - **Guaranteed Convergence:** Always converges (unconditional bracketing).
- **Regula-Falsi Method (False Position):**
  - **Formula:** $x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$.
  - **Order of Convergence (p): $\mathbf{p = 1}$** (Linear, asymptotically slower than Secant due to retained stagnant endpoint).
  - **Error Relation:** $|e_{n+1}| \approx C|e_n|$.
  - **Guaranteed Convergence:** Always converges when bracketed.
- **Secant Method (Open 2-Point Linear Interpolation):**
  - **Formula:** $x_{n+1} = x_n - f(x_n)\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$.
  - **Order of Convergence (p): $\mathbf{p = \frac{1+\sqrt{5}}{2} \approx 1.618}$** (Golden Ratio, Superlinear).
  - **Error Recurrence:** $e_{n+1} \approx C \cdot e_n \cdot e_{n-1} \implies |e_{n+1}| \approx K|e_n|^{1.618}$.
  - **Condition:** No derivative needed; 1 function evaluation per step after start.
- **Newton-Raphson Method (Tangent Method):**
  - **Formula:** $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.
  - **Order of Convergence (p): $\mathbf{p = 2}$** (Quadratic Convergence).
  - **Asymptotic Error Relation:** $e_{n+1} \approx \frac{f''(\alpha)}{2f'(\alpha)}e_n^2$.
  - **Convergence Condition:** Requires $|f(x)f''(x)| < [f'(x)]^2$ near root and $f'(x) \neq 0$.
- **Fixed-Point Iteration Method ($x = \phi(x)$):**
  - **Formula:** $x_{n+1} = \phi(x_n)$.
  - **Order of Convergence (p): $\mathbf{p = 1}$** (Linear).
  - **Sufficient Condition for Convergence:** $|\phi'(x)| \leq k < 1$ for all $x \in [a, b]$.

2. Dedicated Newton-Raphson Dedicated Algorithms

- **Square Root Algorithm ($\sqrt{N}$):**

$$f(x) = x^2 - N = 0 \implies \mathbf{x_{n+1} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right)}$$

- **k-th Root Algorithm ($\sqrt[k]{N}$):**

$$f(x) = x^k - N = 0 \implies \mathbf{x_{n+1} = \frac{1}{k}\left((k-1)x_n + \frac{N}{x_n^{k-1}}\right)}$$

- **Division-Free Reciprocal Algorithm ($1/N$):**

$$f(x) = \frac{1}{x} - N = 0 \implies \mathbf{x_{n+1} = x_n(2 - Nx_n)}$$

  - Initial interval condition: Convergence requires $0 < x_0 < \frac{2}{N}$.

- **Inverse Square Root Algorithm ($1/\sqrt{N}$):**

$$f(x) = \frac{1}{x^2} - N = 0 \implies \mathbf{x_{n+1} = \frac{x_n}{2}(3 - Nx_n^2)}$$

3. Order of Convergence & Derivations Quick Summary

- **Definition of Order of Convergence p:**

$$\lim_{n\to\infty} \frac{|e_{n+1}|}{|e_n|^p} = C \quad (C > 0, p \geq 1)$$

- **Newton-Raphson Error Derivation ($p = 2$):**

$$0 = f(\alpha) = f(x_n) + (\alpha - x_n)f'(x_n) + \frac{(\alpha - x_n)^2}{2!} f''(\xi_n)$$

Dividing by $f'(x_n)$ and using $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$:

$$x_{n+1} - \alpha = \frac{f''(\xi_n)}{2f'(x_n)}(x_n - \alpha)^2 \implies \mathbf{e_{n+1} \approx \frac{f''(\alpha)}{2f'(\alpha)} e_n^2}$$

- **Secant Error Derivation ($p \approx 1.618$):**

$$e_{n+1} \approx C \cdot e_n \cdot e_{n-1} \implies e_n = Ae_{n-1}^p \implies A^{1+p} e_{n-1}^{p^2} \approx CAe_{n-1}^{p+1}$$

Equating exponents gives $p^2 = p + 1 \implies p^2 - p - 1 = 0 \implies \mathbf{p = \frac{1+\sqrt{5}}{2} \approx 1.618}$.

- **Fixed-Point Error Derivation ($p = 1$):**

$$e_{n+1} = x_{n+1} - \alpha = \phi(x_n) - \phi(\alpha) = \phi'(\xi_n)(x_n - \alpha) = \phi'(\xi_n)e_n \implies |\mathbf{e_{n+1}}| \leq \mathbf{k}|\mathbf{e_n}|$$

where $k = \max_{x\in[a,b]} |\phi'(x)| < 1$.